

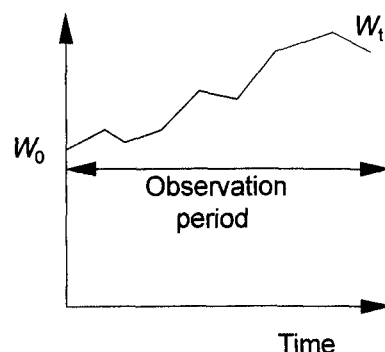


Figure 6.1 A Primer on Time Series.

easy. The pattern of the series would reveal considerable and predictive information. You could tell with precision what would happen one day ahead, one year ahead, and perhaps even a decade ahead. We would not even need a statistician; a second-rate engineer would do. He does not even need to be armed with a modern degree; someone with nineteenth-century training under Laplace would be able to solve the equations, called *differential equations*, or, equivalently, *equations of motion* – since we are studying the dynamics of an entity whose position depends on time.

If we were dealing with a world where randomness is charted, things would be easy as well, given that there is an entire field created for that called *Econometrics* or *Time Series Analysis*. You would call a friendly econometrician (my experience of econometricians is that they are usually polite and friendly to practitioners). He would run the data in his software, and provide you with diagnostics that would tell you if it is worth investing in the trader generating such a track record, or if it is worth pursuing the given trading strategy. You can even buy the student version of his software for under \$999 and run it yourself during the next rainy weekend.

But we are not sure that the world we live in is well charted. We will see that the judgment derived from the analysis of these past attributes may on occasion be relevant. But it may be meaningless; it could on occasion mislead you and take you in the opposite direction. Sometimes market data becomes a simple trap; it shows you the opposite of its nature, simply to get you to invest in the security or mismanage your

risks. Currencies that exhibit the largest historical stability, for example, are the most prone to crashes. This was bitterly discovered in the summer of 1997 by investors who chose the safety of the pegged currencies of Malaysia, Indonesia, and Thailand (they were pegged to the U.S. dollar in a manner to exhibit no volatility, until their sharp, sudden, and brutal devaluations).

We could be either too lax or too stringent in accepting past information as a prediction of the future. As a skeptic, I reject a sole time series of the past as an indication of future performance; I need a lot more than data. My major reason is the *rare event*, but I have plenty of others.

On the surface, my statement here may seem to contradict earlier discussions, where I blame people for not learning enough from history. The problem is that we read too much into shallow recent history, with statements like “this has never happened before”, but not from history in general (things that never happened before in one area tend eventually to happen). In other words history teaches us that things that never happened before do happen. It can teach us a lot outside of the narrowly defined time series; the broader the look, the better the lesson. In other words, history teaches us to avoid the brand of naïve empiricism that consists of learning from casual historical facts.

The Rare Event Fallacy

THE MOTHER OF ALL DECEPTIONS

The rare event, owing to its dissimulative nature, can take a variety of shapes. It is in Mexico that it was spotted first, where it was called by academics the *peso problem*. Econometricians were puzzled by the behavior of the Mexican economic variables during the 1980s. The money supply, interest rates, or some similar measure of small relevance to the story exhibited some moody behavior, thwarting many of their efforts at modeling them. These indicators erratically switched between periods of stability to brief bursts of turbulence without warning.